Cutler-Hammer

Current Cutler-Hammer Panelboards

Renewal Parts

Supersedes RP.38F.01.T.E
pages 1 – 48 dated February 2000

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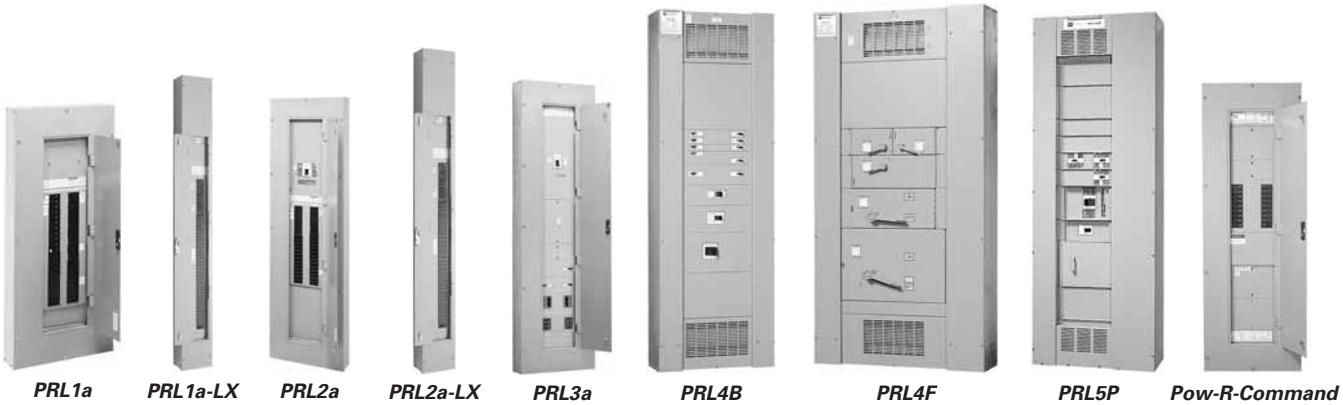

PRL1a *PRL1a-LX* *PRL2a* *PRL2a-LX* *PRL3a* *PRL4B* *PRL4F* *PRL5P* *Pow-R-Command*

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Table 1. Product History Time Line

Product	1985	1990	1995	Present
Cutler-Hammer PRL1a, 2a			←	Oct. 1996 →
Cutler-Hammer PRL3a			←	Mar. 1994 →
Cutler-Hammer PRL4B/F	←			Oct. 1987 →
Cutler-Hammer PRL5P			←	Aug. 1995 →
Cutler-Hammer PRL1a, 2a-LX			←	Dec. 1997 →
Cutler-Hammer Pow-R-Command			←	Mar. 1996 →

Procedure for Identifying Panelboard Type

The current line of Pow-R-line C panelboards was introduced in 1993.

A panelboard is identified by data found on the nameplate. Pow-R-Line C panelboard nameplates are different in appearance, but all have the same critical information:

- Ampere rating of the main.
- Ampere rating of the neutral.
- Type of service (phase/wire).
- Manufacturing location.
- Type of panel.
- General order number.

In the event the nameplate is missing, it may still be possible to identify the panel type by location of the neutral bar. The illustrations to the right shows the position of the neutral in the panelboard.

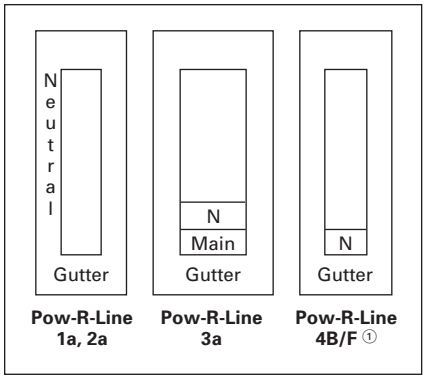

Figure 1. Position of the Neutral in the Panelboard

① PRL4F panels with vertical mounted main switch will have the neutral mounted at the opposite end the main.

Box width may also help identify the panelboard type. Standard width for PRL1a, PRL2a, and PRL3a is 20.00 inches (508.0 mm). PRL4 standard widths are 24.00, 36.00 and 44.00 inches (609.6, 914.4 and 1117.6 mm).

WARNING

HAZARDOUS VOLTAGE WILL CAUSE SEVERE INJURY OR DEATH. TURN OFF POWER SUPPLY TO EQUIPMENT BEFORE WORKING ON IT.

Procedure for Identifying Renewal Parts

1. Identify the type of panelboard, i.e. PRL1a, PRL2a, PRL3a, PRL4, PRL5P by reading the nameplate. Follow the procedure listed to the left.
2. Refer to the listing below and turn to the proper section in this brochure to identify standard parts.

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3. This book identifies those replacement parts most frequently ordered and which are readily available from stock. These parts can be ordered by style or catalog number to speed up processing and delivery.

Distributor Ordering Instructions

1. Specify part by style/part number.
2. Refer to PL01400001E for pricing information. Discount Symbol CE9 applies.
3. Turn to **Page 4** to locate nearest Satellite Plant.
4. Enter the order on the satellite plant via mail, fax or phone.
5. Selling policy 25-000 applies.

Cutler-Hammer Satellite Plants

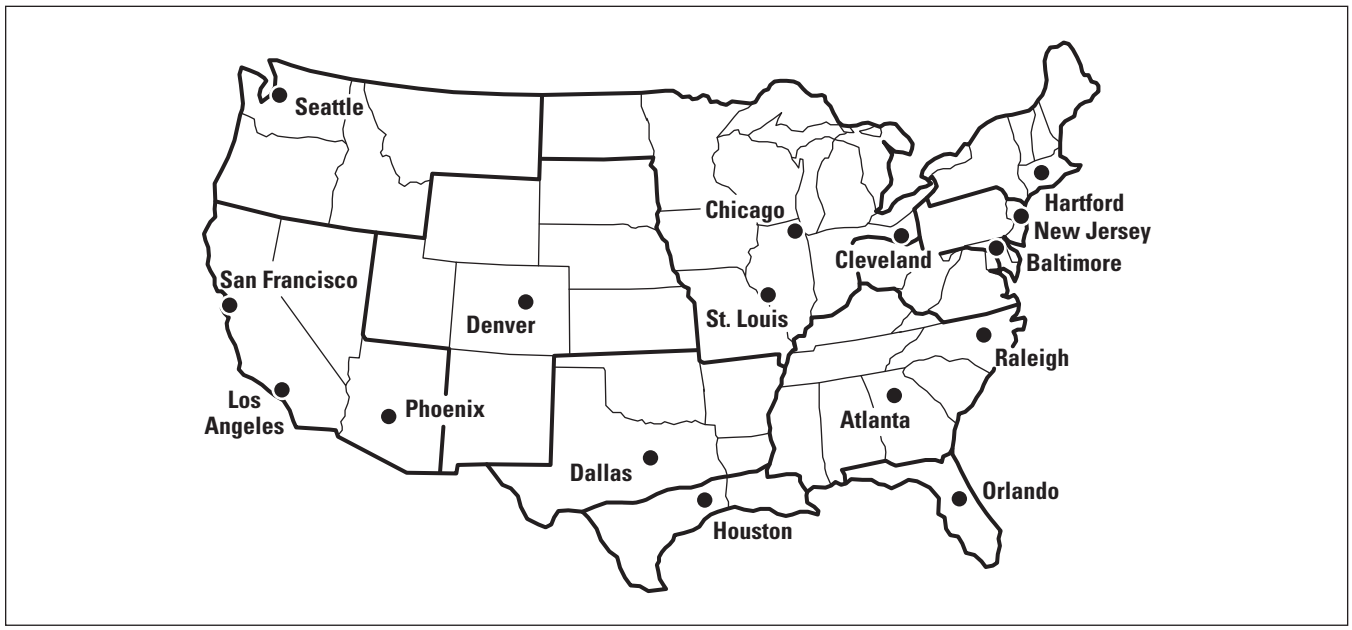

Figure 2. Satellite Plants

Atlanta
7990-A 2nd Flag Drive
Austell, GA 30001
Phone 770-944-1022
FAX 770-944-2033

Baltimore
6671 Santa Barbara Court, Suite A
Elkridge, MD 21227
Phone 410-796-7777
FAX 410-796-7755

Chicago
959 AEC Drive
Wood Dale, IL 60191
Phone 630-860-3500
FAX 630-860-3569

Cleveland
4711 Hinkley Industrial Parkway
Cleveland, OH 44109
Phone 216-485-1940
FAX 216-485-1943

Dallas
1100 Avenue T
Grand Praire, TX 75050
Phone 972-988-3339
FAX 972-641-6435

Denver
14101 East 33rd Place, Suite F
Aurora, CO 80011
Phone: 303-371-7844
FAX 303-371-4175

Hartford
625 Day Hill Road
Windsor, CT 06095
Phone 860-688-7330
FAX 860-688-4982

Houston
10810 West Little York, Suite 100
Houston, TX 77041
Phone 713-688-8430
FAX 713-688-3764

Los Angeles
2021 Locust Court
Ontario, CA 91761
Phone 909-923-2040
FAX 909-923-2344

New Jersey
96 Stemmers Lane
Westampton, NJ 08060
Phone 609-835-4230
FAX 609-835-4777

Orlando
3827 St. Valentine Way
Orlando, FL 32811
Phone 407-843-3863
FAX 407-841-9135

Phoenix
7160 South Harl Avenue
Tempe, AZ 85283
Phone 480-777-3957
FAX 480-777-3958

Raleigh
2933 S. Miami Blvd., Suite 111
Durham, NC 27703
Phone 919-544-7074
FAX 919-572-9751

San Francisco
20919 Cabot Boulevard
Hayward, CA 94545
Phone 510-784-8981
FAX 510-784-8980

Seattle
18657 72nd Avenue South
Kent, WA 98032
Phone 425-251-9081
FAX 425-251-0079

St. Louis
12947 Gravois Road
St. Louis, MO 63127
Phone: 314-842-7797
FAX 314-842-2552

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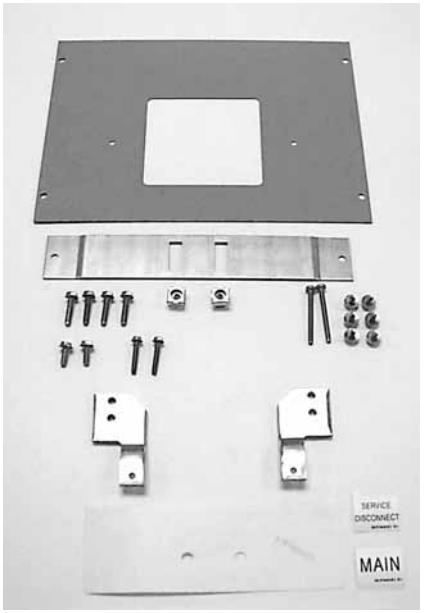
PRL1a, 2a Connector Kits

Table 2. Vertical Breaker Assemblies

Device Type ①	Device Mounting	3-Phase		1-Phase	
		Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
		Catalog Number			
F-Frame ② (100 Ampere Maximum)	Top Fed Bottom Fed	**KB13AFT** **KB13AFB**	**KB13SFT** **KB13SFB**	**KB11AFT** **KB11AFB**	**KB11SFT** **KB11SFB**
F-Frame ③ (225 Ampere Maximum)	Top Fed Bottom Fed	**KB23AFT** **KB23AFB**	**KB23SFT** **KB23SFB**	**KB21AFT** **KB21AFB**	**KB21SFT** **KB21SFB**
J-Frame	Top Fed Bottom Fed	**KB43AJT** **KB43AJB**	**KB43SJT** **KB43SJB**	**KB41AJT** **KB41AJB**	**KB41SJT** **KB41SJB**
K-Frame	Top Fed Bottom Fed	**KB43AKT** **KB43AKB**	**KB43SKT** **KB43SKB**	**KB41AKT** **KB41AKB**	**KB41SKT** **KB41SKB**

① Order main or sub-feed breaker separately when ordering above connector kits.
② EHD, FD, HFD, FDC.
③ FD, HFD, FDC, ED, EDH, EDC.



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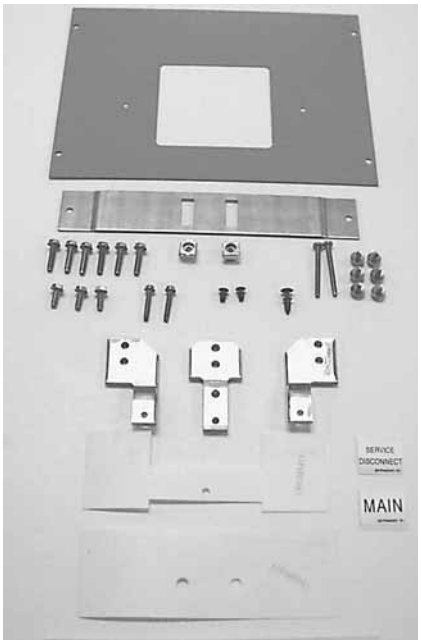


KB13AFT

PRL1a, 2a Connector Kits

Table 3. 100 Ampere Lug Assemblies

Lug Type	Panel Lug Options ①	Wire Size Range	Quantity Per Phase	3-Phase		1-Phase	
				Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
				Catalog Number			
Aluminum/Copper Mechanical	STD SFL OVS	#14 – 1/0 #14 – 1/0 #6 – 300 kcmil	1 2 1	**KL13AMS** **KL13AMF** **KL13AMO**	**KL13SMS** **KL13SMF** **KL13SMO**	**KL11AMS** **KL11AMF** **KL11AMO**	**KL11SMS** **KL11SMF** **KL11SMO**
Crimp	STD SFL OVS	#8 – 1/0 #8 – 1/0 #4 – 300 kcmil	1 2 1	**KL13AVS** **KL13AVF** **KL13AVO**	**KL13SVS** **KL13SVF** **KL13SVO**	**KL11AVS** **KL11AVF** **KL11AVO**	**KL11SVS** **KL11SVF** **KL11SVO**
Copper Mechanical	STD SFL OVS	#14 – 1/0 #14 – 1/0 #6 – 250 kcmil	1 2 1	— — —	**KL13SCS** **KL13SCF** **KL13SCO**	— — —	**KL11SCS** **KL11SCF** **KL11SCO**

① STD = Standard lugs. Use for main or through-feed.
SFL = Sub-feed lugs.
OVS = Oversize lugs. Use for main or through-feed.

KL13AMS

KL11AVS

PRL1a, 2a Connector Kits

Table 4. 225 Ampere Lug Assemblies

Lug Type	Panel Lug Options ①	Wire Size Range	Quantity Per Phase	3-Phase		1-Phase	
				Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
				Catalog Number			
Aluminum/Copper Mechanical	STD SFL OVS	#6 – 300 kcmil #6 – 300 kcmil 4/0 – 500 kcmil	1 2 1	**KL23AMS** **KL23AMF** **KL23AMO**	**KL23SMS** **KL23SMF** **KL23SMO**	**KL21AMS** **KL21AMF** **KL21AMO**	**KL21SMS** **KL21SMF** **KL21SMO**
Crimp	STD SFL OVS	#4 – 300 kcmil #4 – 300 kcmil 2/0 – 500 kcmil	1 2 1	**KL23AVS** **KL23AVF** **KL23AVO**	**KL23SVS** **KL23SVF** **KL23SVO**	**KL21AVS** **KL21AVF** **KL21AVO**	**KL21SVS** **KL21SVF** **KL21SVO**
Copper Mechanical	STD SFL OVS	#6 – 250 kcmil #6 – 250 kcmil 1/0 – 600 kcmil	1 2 1	— — —	**KL23SCS** **KL23SCF** **KL23SCO**	— — —	**KL21SCS** **KL21SCF** **KL21SCO**

① STD = Standard lugs. Use for main or through-feed.
SFL = Sub-feed lugs.
OVS = Oversize lugs. Use for main or through-feed.

KL23AMS

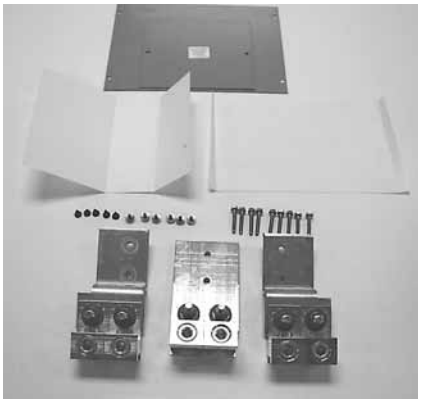
KL21AVS

PRL1a, 2a Connector Kits

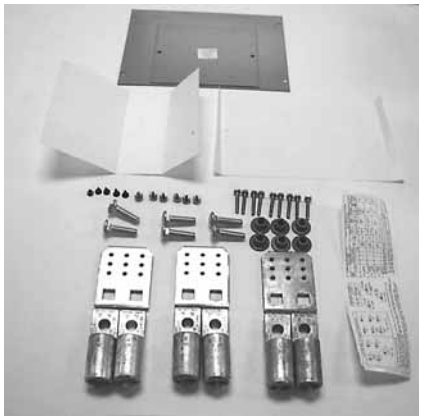
Table 5. 400 Ampere Lug Assemblies

Lug Type	Panel Lug Options ①	Wire Size Range	Quantity Per Phase	3-Phase		1-Phase	
				Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
				Catalog Number			
Aluminum/Copper Mechanical	STD SFL OVS	4/0 – 500 kcmil N/A 3/0 – 750 kcmil	2 N/A 2	**KL43AMS** **—** **KL43AMO**	**KL43SMS** **—** **KL43SMO**	**KL41AMS** **—** **KL41AMO**	**KL41SMS** **—** **KL41SMO**
Crimp	STD SFL OVS	2/0 – 500 kcmil N/A 500 – 750 kcmil	2 N/A 2	**KL43AVS** **—** **KL43AVO**	**KL43SVS** **—** **KL43SVO**	**KL41AVS** **—** **KL41AVO**	**KL41SVS** **—** **KL41SVO**
Copper Mechanical	STD SFL OVS	1/0 – 600 kcmil N/A 1/0 – 600 kcmil	1 N/A 1	**—** **—** **—**	**—** **—** **KL43SCO**	**—** **—** **—**	**—** **—** **KL41SCO**

① STD = Standard lugs. Use for main or through-feed.
SFL = Sub-feed lugs.
OVS = Oversize lugs. Use for main or through-feed.



KL43AMS



KL43AVS

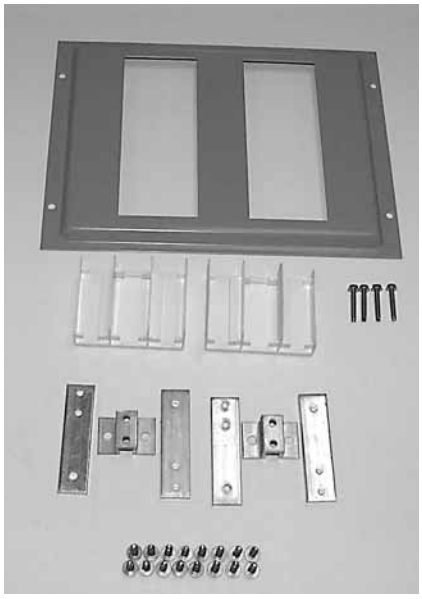
PRL1a Horizontally Mounted Connector Kit Assemblies

Table 6. Bolt-on QUICKLAG Breaker Assemblies

Breaker Frame	Drawing Number ①	Branch Circuit Quantity	3-Phase		1-Phase	
			Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
			Item Number			
BA, BAB, QBH, QBGF, QBHGF, QBGFEP, QBHGFEP	1C96608	12 18 30	G01 G09 G17	G03 G11 G19	G05 G13 G21	G07 G15 G23
		42 48 54	G25 G33 G41	G27 G35 G43	G29 G37 G45	G31 G39 G47
		72 96	G49 G57	G51 G59	G53 G61	G55 G63

① Order the basic drawing number, along with the equivalent G–number that's needed.

Note: When determining branch circuit quantity, remember:
1. QUICKLAG breakers with shunt trips require one additional circuit.
2. UL® listed lighting and appliance (CTL) panelboards **cannot** exceed 42 electrically connected circuits in a single enclosure.
3. When bare copper is specified, use the silver-plated groups.
4. **Order breakers separately with connector kit.**



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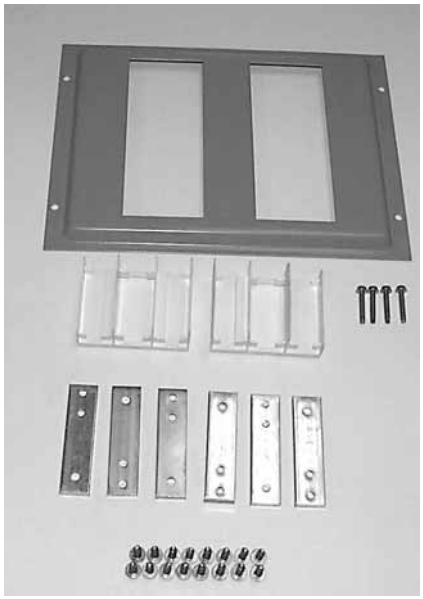

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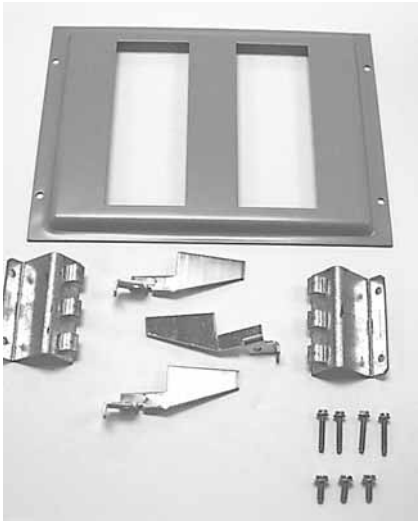
PRL1a Horizontally Mounted Connector Kit Assemblies

Table 7. Plug-in QUICKLAG Breaker Assemblies

Breaker Frame	Drawing Number ①	Branch Circuit Quantity	3-Phase		1-Phase	
			Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
			Item Number			
HQP, QPHW, QHPX, QPGF, QPHGF, QPGFEP, QPHGFEP	2C11642	12	—	G03	—	G07
		18	—	G11	—	G15
		30	—	G19	—	G23
		42	—	G27	—	G31
		48	—	G35	—	G39
		54	—	G43	—	G47
		72	—	G51	—	G55
		96	—	G59	—	G63

① Order the basic drawing number, along with the equivalent G–number that's needed.

Note: When determining branch circuit quantity, remember:
1. QUICKLAG breakers with shunt trips require one additional circuit.
2. UL listed lighting and appliance (CTL) panelboards **cannot** exceed 42 electrically connected circuits in a single enclosure.
3. When aluminum is specified, use the silver-plated groups.
4. The sum of the horizontally twin mounted breakers **shall not exceed 140 amperes**.
5. **Order breakers separately with connector kit**.



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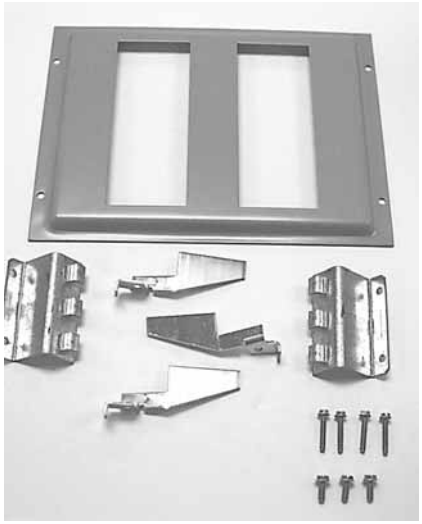


2C11642G07

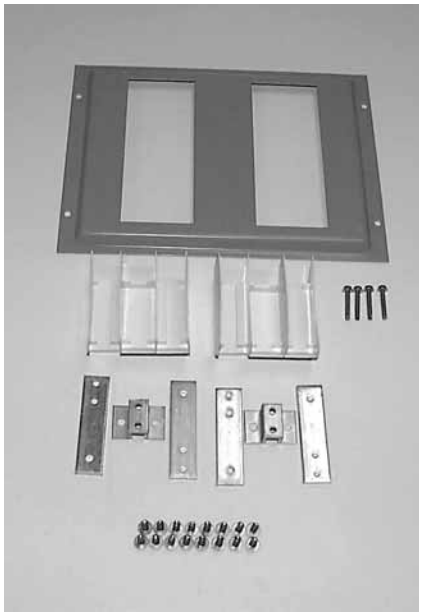
PRL2a Horizontally Mounted Connector Kit Assemblies

Table 8. GB, GHB, GHQ, GHBS Breaker Assemblies

Breaker Frame	Drawing Number ①	Branch Circuit Quantity	3-Phase		1-Phase	
			Tin-Plated Aluminum Connector	Silver-Plated Copper Connector	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
			Item Number			
GB, GHB, GHQ GHBS	1C96609	12 18 30	G01 G09 G17	G03 G11 G19	G05 G13 G21	G07 G15 G23
		42 48 54	G25 G33 G41	G27 G35 G43	G29 G37 G45	G31 G39 G47
		72 96	G49 G57	G51 G59	G53 G61	G55 G63

① Order the basic drawing number, along with the equivalent G–number that's needed.

Note: When determining branch circuit quantity, remember:
1. QUICKLAG breakers with shunt trips require one additional circuit.
2. UL listed lighting and appliance (CTL) panelboards **cannot** exceed 42 electrically connected circuits in a single enclosure.
3. When bare copper is specified, use the silver-plated groups.
4. **Order breakers separately with connector kit.**



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PRL1a, 2a Neutral Assemblies

Table 9. 100 Ampere Neutral Assemblies ①

Panel Main Bus Ampere Rating	Neutral Rating	Lug Type	Drawing Number ②	Panel Lug Options ③	Wire Size Range	Quantity	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
							Item Number	
100	100%	Mechanical	1C96646	STD SFL/TFL OVS	#14 – 1/0 #14 – 1/0 #6 – 300 kcmil	1 2 1	G02 G05 G09	G03 G07 G11
		Crimp	1C96647	STD SFL/TFL OVS	#8 – 1/0 #8 – 1/0 #4 – 300 kcmil	1 2 1	G01 G05 G09	G03 G07 G11
		Copper	1C96648	STD SFL/TFL OVS	#14 – 1/0 #14 – 1/0 #6 – 250 kcmil	1 2 1	— — —	G03 G07 G11
	200%	Mechanical	1C96649	STD SFL/TFL OVS	#6 – 300 kcmil #6 – 300 kcmil 4/0 – 500 kcmil	1 2 1	G02 G06 G09	G03 G07 G11
		Crimp	1C96650	STD SFL/TFL OVS	#4 – 300 kcmil #4 – 300 kcmil 2/0 – 500 kcmil	1 2 1	G01 G05 G09	G03 G07 G11
		Copper	1C96651	STD SFL/TFL OVS	#6 – 250 kcmil #6 – 250 kcmil 1/0 – 600 kcmil	1 2 1	— — —	G03 G07 G11

① The assemblies shown on this page are for panelboards that mount in 30.00 – 90.00-inch (762.0 – 2286.0 mm) high enclosures only. Reference **Page 15** for assemblies for panelboards that mount in 21.00 – 27.00-inch (533.4 – 685.8 mm) high enclosures.

② Order the basic drawing number, along with the equivalent G–number that's needed.

③ STD = Standard lugs.
SFL/TFL = Sub-feed and through-feed lugs.
OVS = Oversize lugs.

1C96646G01

PRL1a, 2a Neutral Assemblies

Table 10. 225 Ampere Neutral Assemblies ①

Panel Main Bus Ampere Rating	Neutral Rating	Lug Type	Drawing Number ②	Panel Lug Options ③	Wire Size Range	Quantity	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
							Item Number	
225	100%	Mechanical	1C96649	STD SFL/TFL OVS	#6 – 300 kcmil #6 – 300 kcmil 4/0 – 500 kcmil	1 2 1	G02 G06 G09	G03 G07 G11
		Crimp	1C96650	STD SFL/TFL OVS	#4 – 300 kcmil #4 – 300 kcmil 2/0 – 500 kcmil	1 2 1	G01 G05 G09	G03 G07 G11
		Copper	1C96651	STD SFL/TFL OVS	#6 – 250 kcmil #6 – 250 kcmil 1/0 – 600 kcmil	1 2 1	— — —	G03 G07 G11
	200%	Mechanical	1C96652	STD SFL/TFL OVS	4/0 – 500 kcmil N/A 3/0 – 750 kcmil	2 N/A 2	G01 G05 G09	G03 G07 G11
		Crimp	1C96653	STD SFL/TFL OVS	2/0 – 500 kcmil N/A 500 – 750 kcmil	2 N/A 2	G01 G05 G09	G03 G07 G11
		Copper	1C96654	STD SFL/TFL OVS	1/0 – 600 kcmil N/A 1/0 – 600 kcmil	1 N/A 1	— — —	G03 G07 G11

① The assemblies shown on this page are for panelboards that mount in 30.00 – 90.00-inch (762.0 – 2286.0 mm) high enclosures.
② Order the basic drawing number, along with the equivalent G–number that's needed.
③ STD = Standard lugs.
SFL/TFL = Sub-feed and through-feed lugs.
OVS = Oversize lugs.

1C96649G01

PRL1a, 2a Neutral Assemblies

Table 11. 400 Ampere Neutral Assemblies ①

Panel Main Bus Ampere Rating	Neutral Rating	Lug Type	Drawing Number ②	Panel Lug Options ③	Wire Size Range	Quantity	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
							Item Number	
400	100%	Mechanical	1C96652	STD SFL/TFL OVS	4/0 – 500 kcmil N/A 3/0 – 750 kcmil	2 N/A 2	G01 G05 G09	G03 G07 G11
		Crimp	1C96653	STD SFL/TFL OVS	2/0 – 500 kcmil N/A 500 – 750 kcmil	2 N/A 2	G01 G05 G09	G03 G07 G11
		Copper	1C96654	STD SFL/TFL OVS	1/0 – 600 kcmil N/A 1/0 – 600 kcmil	1 N/A 1	— — —	G03 G07 G11

① The assemblies shown on this page are for panelboards that mount in 30.00 – 90.00-inch (762.0 – 2286.0 mm) high enclosures.
② Order the basic drawing number, along with the equivalent G–number that's needed.
③ STD = Standard lugs.
SFL/TFL = Sub-feed and through-feed lugs.
OVS = Oversize lugs.

1C96652G01

PRL1a, 2a Neutral Assemblies

Table 12. 100 Ampere Neutral Assemblies for 21.00 – 27.00-Inch (533.4 – 685.8 mm) High Enclosures Only ①

Panel Main Bus Ampere Rating	Neutral Rating	Lug Type	Drawing Number ②	Panel Lug Options ③	Wire Size Range	Quantity	Tin-Plated Aluminum Connector	Silver-Plated Copper Connector
							Item Number	
100	100%	Mechanical	1C96645	STD SFL/TFL OVS	#14 – 1/0 #14 – 1/0 N/A	1 2 N/A	G01 G05 —	G03 G07 —
		Crimp	N/A	STD SFL/TFL OVS	N/A N/A N/A	N/A N/A N/A	— — —	— — —
		Copper	N/A	STD SFL/TFL OVS	N/A N/A N/A	N/A N/A N/A	— — —	— — —
	200%	Mechanical	1C97022	STD SFL/TFL OVS	#6 – 300 kcmil #6 – 300 kcmil N/A	1 2 N/A	G01 G05 —	G03 G07 —
		Crimp	N/A	STD SFL/TFL OVS	N/A N/A N/A	N/A N/A N/A	— — —	— — —
		Copper	N/A	STD SFL/TFL OVS	N/A N/A N/A	N/A N/A N/A	— — —	— — —

① The assemblies shown on this page are for panelboards that mount in 21.00 – 27.00-inch (533.4 – 685.8 mm) high enclosures only. Reference **Page 12** for assemblies for panels that mount in 36.00, 48.00, 60.00, 72.00 and 90.00-inch (914.4, 1219.2, 1524.0, 1828.8 and 2286.0 mm) high enclosures.
② Order the basic drawing number, along with the equivalent G–number that's needed.
③ STD = Standard lugs.
SFL/TFL = Sub-feed and through-feed lugs.
OVS = Oversize lugs.

1C96645G01

PRL1a, 2a Ground Assemblies

Table 13. Standard Ground

Drawing Number ①	Enclosure Height in Inches (mm)	Bar Material	Item Number
5158C05	24.00 (609.6)	Aluminum/Copper	G01
		Copper	G03
	36.00 (914.4), 48.00 (1219.2), 60.00 (1524.0), 72.00 (1828.8), 90.00 (2286.0)	Aluminum/Copper	G02
		Copper	G04

① Order the basic drawing number, along with the equivalent G–number that's needed (example 5158C05G01).

5158C05G01

5158C05G02

Table 14. Isolated Ground

Drawing Number ②	Enclosure Height in Inches (mm)	Bar Material	Item Number
2C11296	24.00 (609.6)	Aluminum/Copper	G01
		Copper	G02
	36.00 (914.4), 48.00 (1219.2), 60.00 (1524.0), 72.00 (1828.8), 90.00 (2286.0)	Aluminum/Copper	G03
		Copper	G04

② Order the basic drawing number, along with the equivalent G–number that's needed (example 5158C05G01).

PRL1a, 2a Service Entrance Kits

Table 15. PRL1a, 2a Service Entrance Kits

Drawing Number ③	Panel Ampere Rating	Tin-Plated Aluminum	Bare Copper	Silver-Plated Copper	Tin-Plated Copper
		Item Number			
Mechanical Main Lugs or Main Breakers					
4180B62	100 – 225	G01	G02	G03	G04
4180B62	400	G05	G06	G07	G08
Compression (Crimp) Main Lugs					
4180B62	100 – 225	G09	G10	G11	G12
4180B62	400	G13	G14	G15	G16
Copper Main Lugs					
4180B62	100 – 225	—	G18	G19	G20
4180B62	400	—	G22	G23	G24

③ Order the basic drawing number, along with the equivalent G–number that's needed (example 5158C05G01).

4180B62G01

PRL1a, 2a Deadfront Covers

Note: Does not apply to PRL4 sub-chassis.

1C96638G01

Table 16. Assembly

Drawing Number ①	Standard Enclosure Height in Inches (mm)						
	24.00 (609.6)	36.00 (914.4)	42.00 (1066.8)	48.00 (1219.2)	60.00 (1524.0)	72.00 (1828.8)	90.00 (2286.0)
1C96638	G01	G02	G07	G03	G04	G05	G06

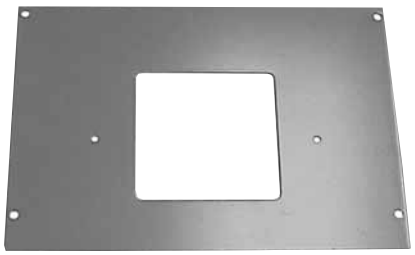
① Order the basic drawing number, along with the equivalent G–number that's needed (example 1C96638G01).

Table 17. Vertically Mounted Devices

Mounting Arrangement	Device/Frame	Drawing Number ②	Mounting Position	Item Number
Vertical	100 Ampere MLO, SFL, TFL or F-Frame (100 Ampere Maximum)	4180B03	Top Bottom	H01 H01
	225 Ampere MLO, SFL, TFL or F-Frame (225 Ampere Maximum)	4180B61	Top Bottom	H01 H01
	400 Ampere MLO, SFL, TFL or J-Frame	4180B04	Top Bottom	H01 H02
	400 Ampere MLO, TFL or K-Frame	4180B05	Top Bottom	H01 H02
Blank Covers in Inches (mm)	1.00 (25.4) 2.00 (50.8) 3.00 (76.2)	4180B08	N/A N/A N/A	H01 H02 H03
	4.00 (101.6) 5.00 (127.0) 6.00 (152.4)		N/A N/A N/A	H04 H05 H06
	7.00 (177.8) 8.00 (203.2) 9.00 (228.6)		N/A N/A N/A	H07 H08 H09
	10.00 (254.0) 11.00 (279.4) 12.00 (304.8)		N/A N/A N/A	H10 H11 H12
	13.00 (330.2) 14.00 (355.6) 15.00 (381.0) 16.00 (406.4)		N/A N/A N/A N/A	H13 H14 H15 H16

② Order the basic drawing number, along with the equivalent H–number that's needed (example 4180B03H01).

4180B08H03



4180B03H01

PRL1a, 2a Deadfront Covers

Note: Does not apply to PRL4 sub-chassis.

Table 18. Horizontally Mounted Devices

Mounting Arrangement	Device/ Frame	Drawing Number ①	Branch Circuit Quantity	Item Number	Quantity Required
Horizontal	BA, BAB, QBH, QBGF, QBHGF, QBGFEP, QBHGFEP	1C96619	12 18 30	H01 H02 H04	1 1 1
			42 48 54	H06 H03 H03 and H04	1 2 1 Each
			72 96	H05 H07	2 2
	GB, GHB, GHQ, GHBS	1C96620	12 18 30	H01 H02 H04	1 1 1
			42 48 54	H06 H03 H03 and H04	1 2 1 Each
			72 96	H05 H07	2 2

① Order the basic drawing number, along with the equivalent H–number that's needed (example 1C96619H01).

Table 19. Filler Covers

Device/Frame	Drawing Number	Item Number
F, J, K ②	4180B52	H01
QUICKLAG, GB, GHB ③	5155C62	H01

② Filler covers are required in addition to deadfront cover whenever MLO, SFL or TFL are specified.

③ Filler covers are required in addition to deadfront cover whenever a branch provision is specified.

1C96619H01

1C96620H01

5155C62H01

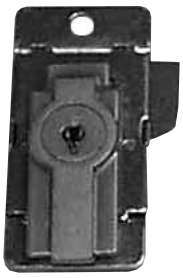
4180B52H01

Panelboard Trim Locks

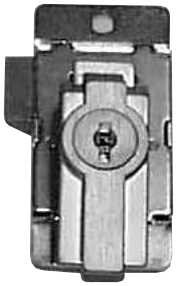
Panelboard trims use different trim locks, see pictures below for styles and part numbers. Contact your nearest Satellite for availability on the styles listed below. See **Page 4** for Satellite listings.

Table 20. Panelboard Trim Locks

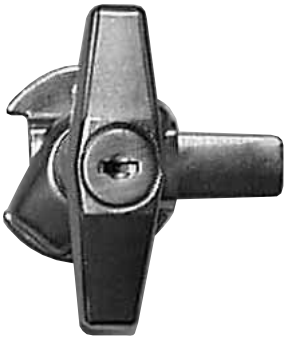
Description	Catalog Number
For use on left-handed door. (Hinged on left side.)	**K80522**
For use on right-handed door. (Hinged on right side.)	**K80133**
T–Handle lock, at one time used on all trims over 48.00 inches (1219.2 mm) in height. Also used on outdoor NEMA 12/3R trims.	**K80429**
Used on PRL4 lighting and power panels as standard.	**1A32258H03**
Used on PRL1, 2, 3 and PRL1a, 2a, 3a lighting panels as standard. WEM 2 key.	**5155C81G01**



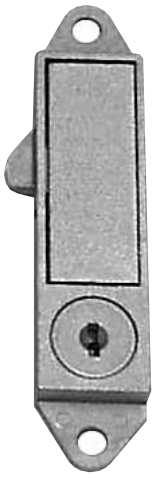
K80522



K80133



K80429



1A32258H03

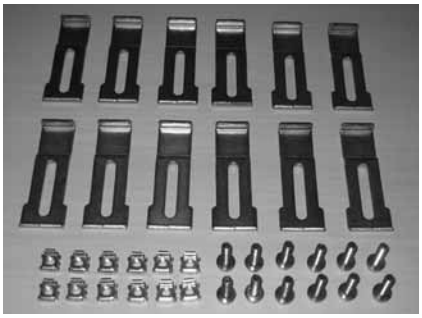
5155C81G01

Panelboard Fastrim Clamps and Screw-on Hardware Kits

For panelboard trim clamps, contact your nearest Satellite for availability on the styles listed below. See **Page 4** for Satellite listings.

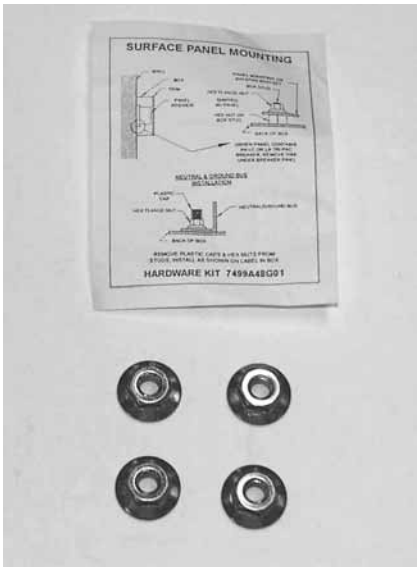
Table 21. Panelboard Fastrim Clamps and Screw-on Hardware Kits

Description	Style Number
Trim clamps — used on PRL1a, 2a, 3a fastrims. (6 per bag.)	**2C11641G02**
Trim screws — used on PRL1a, 2a, 3a, 4B standard trim. (10 per bag.)	**5157C83G06**
Chassis mounting hardware bag — PRL1a, 2a, 3a panels.	**7499A48G04**



2C11641G02

5157C83G06



7499A48G04

PRL3a Parts Section *Page*

Connector Kits, Branch Breakers **21**
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GB, GHB, GHBS **21**
Twin Mounted F-Frame 150 Ampere Maximum **22**
Single Mounted F-Frame 175 – 225 Ampere Maximum . **22**
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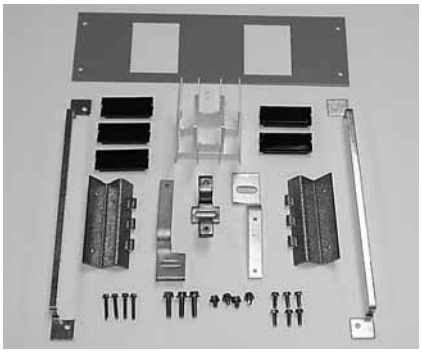
PRL3a Horizontally Mounted Connector Kit Assemblies

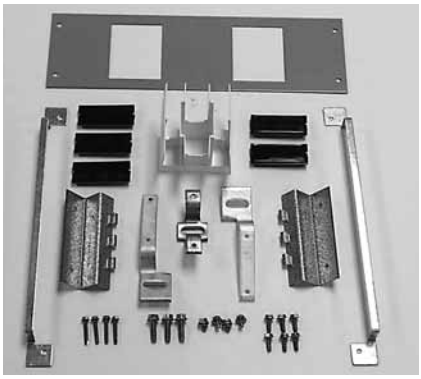
Table 22. Connector Kit Assemblies

Devices	Circuits or Pole	3-Phase		1-Phase		Notes
		Catalog Number	Phase	Catalog Number	Phase	
BA, BAB, QBGF, QBH, QBHGF, QBGFEP, QBHGFEP	6	**KPRL3ABA06**	A/B/C	**KPRL3ABA06-1**	A/C	(2) 100 Ampere Devices Maximum
	12	**KPRL3ABA12**	A/B/C	**KPRL3ABA12-1**	A/C	
	18	**KPRL3ABA18**	A/B/C	**KPRL3ABA18-1**	A/C	
	24	**KPRL3ABA24**	A/B/C	**KPRL3ABA24-1**	A/C	
GB, GHB, GHQ, GHBS	6	**KPRL3AGB06**	A/B/C	**KPRL3AGB06-1**	A/C	
	12	**KPRL3AGB12**	A/B/C	**KPRL3AGB12-1**	A/C	
	18	**KPRL3AGB18**	A/B/C	**KPRL3AGB18-1**	A/C	
	24	**KPRL3AGB24**	A/B/C	**KPRL3AGB24-1**	A/C	

Three-phase kits contain A, B and C phase connectors. Single-phase kits contain A and C phase connectors, deadfront cover, hardware and instructions to twin mount breakers across from each other. **Maximum amperes connected to any one connector cannot exceed 200 amperes.**



KPRL3ABA06



KPRL3AGB06

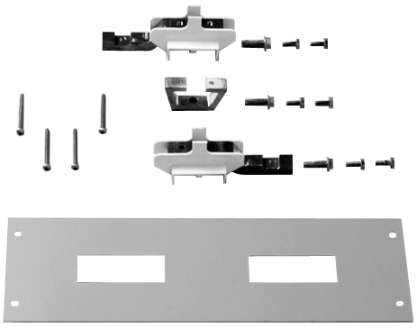
PRL3a F-Frame Horizontally Mounted Connector Kit Assemblies

Table 23. Connector Kit Assemblies

Devices	Circuits or Pole	3-Phase		1-Phase		Notes
		Catalog Number	Phase	Catalog Number	Phase	
EHD, FD, FDB, HFD, FDC (150 Ampere Maximum Twin Mount)	3-Pole Breaker 2-Pole Breaker 1-Pole Breaker	**KPRL3AFD3** **KPRL3AFD2** **KPRL3AFD1**	A/B/C A/C A/C	**—** **KPRL3AFD2** **KPRL3AFD1**	— A/C A/C	(2) 150 Ampere Devices Maximum
FD, HFD, FDC, ED, EDH, EDC (175 – 225 Ampere Single Mount) ①	3-Pole Breaker 2-Pole Breaker	**KPRL3AED3** **KPRL3AED2**	A/B/C A/C	**—** **KPRL3AED2**	— A/C	(1) 225 Ampere Maximum Single Mounted

① F-Frame devices rated above 150 amperes must be single mounted. No twin mounting acceptable.

Connector kits contain phase connectors, deadfront cover, hardware and instructions to mount breakers. Order breakers separately when ordering connector kit.



KPRL3AFD3

PRL3a Ground Assemblies

Table 24. PRL3a Ground Assemblies

Material	Standard	Isolated
	Catalog Number	
Aluminum/Copper Copper Only	**5158C05G02** **5158C05G04**	**2C11296G02** **2C11296G04**

5158C05G02

PRL3a Service Entrance Kits

Table 25. PRL3a Service Entrance Kits

Style Number ①	Panel Ampere Rating	Tin-Plated Aluminum	Bare Copper	Silver-Plated Copper	Tin-Plated Copper
		Item Number			
Mechanical Main Lugs or Main Breakers					
5078A98	100 250 – 600	G01 G13	G02 G14	G03 G15	G04 G16
Crimp Main Lugs					
5078A98	100 250 – 600	G05 G17	G06 G18	G07 G19	G08 G20
Copper Main Lugs					
5078A98	100 250 – 600	G09 G21	G10 G22	G11 G23	G12 G24

① When ordering, use complete style number (example 100 Ampere Tin-Plated Aluminum 5078A98G01).

5078A98G01

PRL3a Deadfront Covers

Table 26. Assembly ②

Style Number ③	Chassis Height/Item Number				
	14X	23X	31X	40X	53X
6559C59	G01	G02	G03	G04	G05

② Assembly groups include the frame only (two rails and two end covers). Reference **Pages 24** and **25** for specific device covers. All connector kits ship with a deadfront cover for that device.

③ When ordering, use complete style number (example 14X High Assembly 6559C59G01).

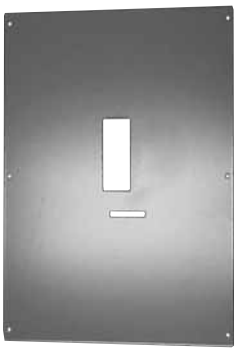
6559C59G01

PRL3a Vertical Devices Deadfront Covers

Table 27. Vertical Mounting Position

Device/Frame	Trip Unit Type	Style Number ①	"X" Space Required	Item Number	
				Without Lock-offs ①	With Lock-offs ①
EHD, FD, FDB, HFD, FDC, ED, EDH, EDC (Top) ②	N/A	**4176B68**	7X	H01	H03
EHD, FD, FDB, HFD, FDC, ED, EDH, EDC (Bottom) ②	N/A		7X	H04	H05
FD, HFD, FDC, ED, EDH (Top) ③	N/A	**4180B93**	10X	H01	H03
FD, HFD, FDC, ED, EDH (Bottom) ③				H04	H05
J-Frame (Bottom)	N/A	**4176B60**	14X	H01	H02
J-Frame (Top)			14X	H03	H04
K-Frame (Bottom)	Thermal-Mag.	**4176B61**	15X	H01	H02
K-Frame (Bottom)	Electronic		15X	H03	H04
K-Frame (Top)	Thermal-Mag.		15X	H05	H06
K-Frame (Top)	Electronic		15X	H07	H08
L-Frame (Bottom)	Thermal-Mag.	**4176B51**	17X	H01	H02
L-Frame (Bottom)	Electronic		17X	H03	H04
L-Frame (Top)	Thermal-Mag.		17X	H05	H06
L-Frame (Top)	Electronic		17X	H07	H08
FB-P (Top Only)	N/A	**4176B70**	9X	H02	H02
LA-P (Top Only)	N/A	**4176B57**	21X	H01	H01
FCL	N/A	**4176B70**	9X	H01	H01
LCL (Top)	N/A	**4176B56**	21X	H01	H02
LCL (Bottom)	N/A		21X	H03	H04
Neutral/Blank Cover	N/A	**4176B72**	1X	H01	—
			2X	H02	
			3X	H03	
			4X	H04	
			5X	H05	
			6X	H06	
			7X	H07	
			8X	H08	
			9X	H09	
			10X	H10	
			11X	H11	
			12X	H12	
J-Frame Sub-Feed Twin Bottom	N/A	**4176B79**	20X	H01	H02 (2 L/O) H03 (1 L/O RT) H04 (1 L/O LT)
J-Frame Sub-Feed Twin Top	N/A	**4176B79**	20X	H05	H05 (2 L/O) H07 (1 L/O RT) H08 (1 L/O LT)
PT363 (Top)	N/A	**4180B79**	7X	H01	—
PT363 (Bottom)	N/A		7X	H02	
PT364 (Top)	N/A		9X	H03	
PT364 (Bottom)	N/A		9X	H04	

① When ordering covers, order complete style and item numbers (example 4176B68H01).
② 4/0 Maximum acceptable terminal size.
③ 300 kcmil maximum acceptable terminal size.



J Main 4176B60H04

Neutral Blank Cover 4176B72H04

PRL3a Horizontal Devices Deadfront Covers

Table 28. Horizontal Mounting Position

Device/Frame	Device Poles	Style Number ①	Total Circuit Quantity	"X" Space Required	Item Number
EHD, FD, FDB, HFD, FDC (Twin Mounted)	1, 2 or 3	**4178B08**	6 12 18 24 30 36 42 48	3X 6X 9X 12X 15X 18X 21X 24X	H01 H02 H03 H04 H05 H06 H07 H08
EHD, FD, FDB, HFD, FDC (Twin Mounted)	1 or 2	**4179B39**	4 8 12 16 20 24 28 32	2X 4X 6X 8X 10X 12X 14X 16X	H01 H02 H03 H04 H05 H06 H07 H08
EHD, FD, FDB, HFD, FDC (Twin Mounted)	1	**4179B40**	2	1X	H01
FD, HFD, FDC, ED, EDH, EDC (Single Mounted)	3	**4179B41**	3	3X	H01
FD, HFD, FDC, ED, EDH, EDC (Single Mounted)	2	**4179B42**	2	2X	H01
CA, CAH, HCA	3	**4176B66**	3	3X	H01
CA, CAH, HCA	2	**4176B80**	2	2X	H01
BA, BAB, BABRP, BABRSP QBH, QBGF, QBGFEP, QBHGFEP	1, 2 or 3	**4176B67**	6 12 18 24	3X 5X 8X 10X	H01 H02 H03 H04
GB, GHB, GHBS, GHBGFEP, HGHB, GHQ	1, 2 or 3	**4176B69**	6 12 18 24	3X 5X 8X 10X	H01 H02 H03 H04
Pow-R-Command Controller	N/A	**4180B91**	N/A	5X	H01
Pow-R-Command Expansion	N/A	**4180B91**	N/A	7X 16X	H02 H03

① When ordering covers, order complete style and item number (example 4178B08H01).

PRL3a Deadfront Cover Blank Fillers

Table 29. PRL3a Deadfront Cover Blank Fillers

Device/Frame	Poles	Style Number
F-Frame C-Frame C-Frame	1, 2 or 3 2 3	**4178B06H01** **6555C40H01** **6555C41H01**
QUICKLAG, GB, GHB, GHBS	1, 2 or 3	**5155C62H01**

BAB Cover 4176B67H01

PRL4B/F Parts Section ***Page***

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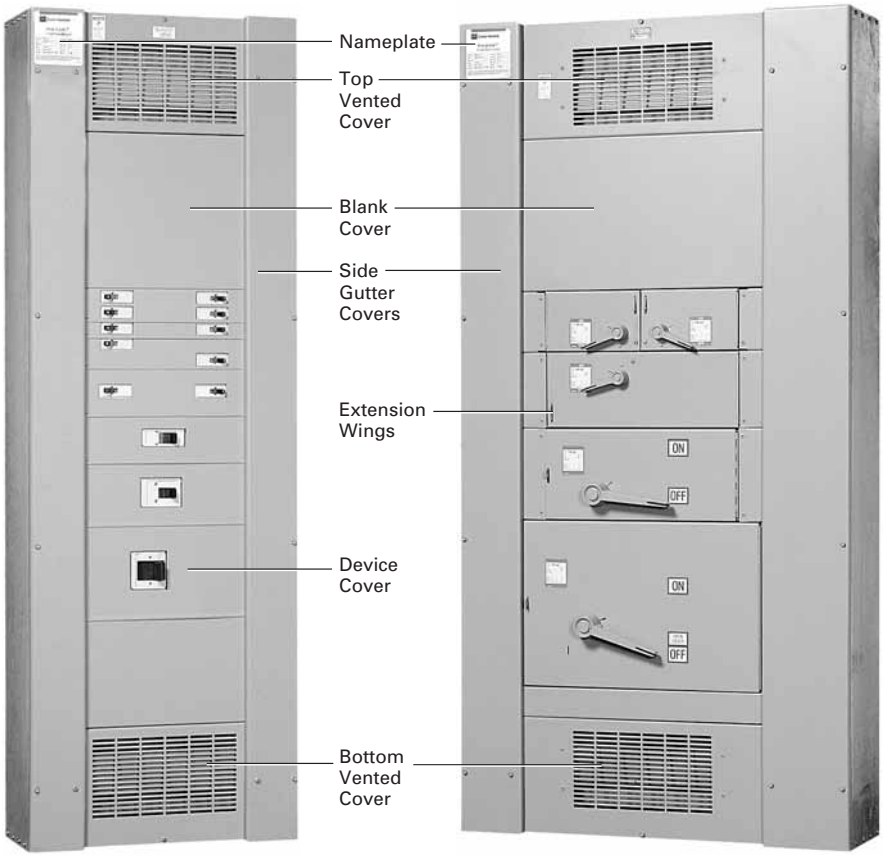
PRL4 Vented Cover Assemblies



PRL4B ***PRL4F***

Table 30. Vented Cover Assemblies and Side Gutter Covers — Dimensions in Inches (mm)

NEMA 1 Box				Vented Cover Assembly ② Style Number	Side Gutter Covers			
Dimensions			Catalog Number		Left		Right	
Height	Width	Depth ①			Size	Style Number	Size	Style Number
57.00 (1447.8)	24.00 (609.6)	10.40 (264.2)	**BX2457**	**6574C74G02**	5.00 (127.0) x 57.00 (1447.8)	**6555C20H01**	5.00 (127.0) x 57.00 (1447.8)	**6555C20H01**
73.00 (1854.2)	24.00 (609.6)		**BX2473**	**6574C74G03**	5.00 (127.0) x 73.00 (1854.2)	**6555C21H01**	5.00 (127.0) x 73.00 (1854.2)	**6555C21H01**
90.00 (2286.0)	24.00 (609.6)		**BX2490**	**6574C74G04**	5.00 (127.0) x 90.00 (2286.0)	**6555C25H01**	5.00 (127.0) x 90.00 (2286.0)	**6555C25H01**
73.00 (1854.2)	36.00 (914.4)		**BX3673**	**6574C74G05**	6.00 (152.4) x 73.00 (1854.2)	**6555C22H01**	8.00 (203.2) x 73.00 (1854.2)	**6555C23H01**
90.00 (2286.0)	36.00 (914.4)		**BX3690**	**6574C74G06**	6.00 (152.4) x 90.00 (2286.0)	**6555C26H01**	8.00 (203.2) x 90.00 (2286.0)	**6555C27H01**
73.00 (1854.2)	44.00 (1117.6)		**BX4473**	**6574C74G05**	8.00 (203.2) x 73.00 (1854.2)	**6555C23H01**	14.00 (355.6) x 73.00 (1854.2)	**6555C24H01**
90.00 (2286.0)	44.00 (1117.6)		**BX4490**	**6574C74G06**	8.00 (203.2) x 90.00 (2286.0)	**6555C27H01**	14.00 (355.6) x 90.00 (2286.0)	**6555C28H01**

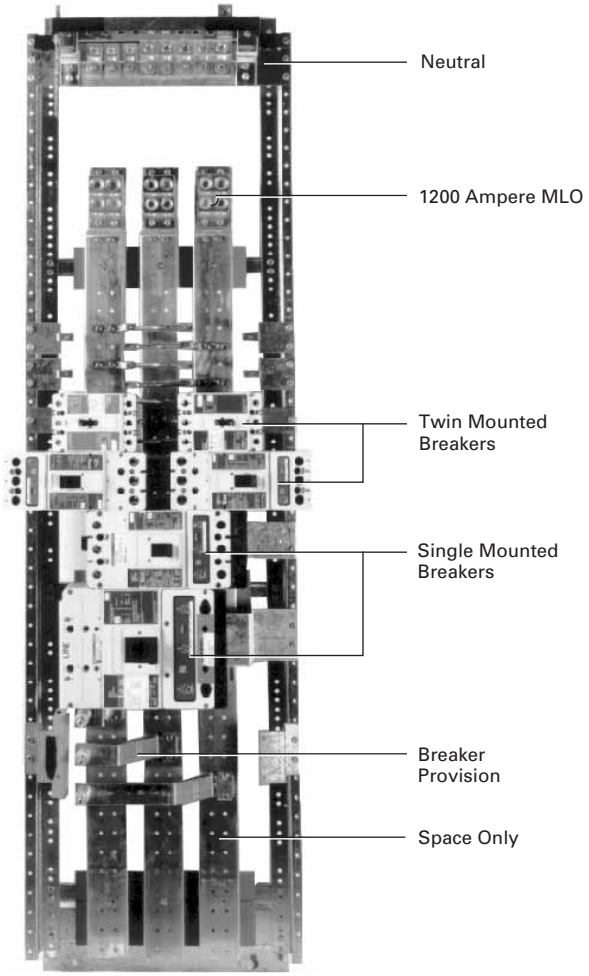
① Covers add .90 inches (22.9 mm) to box depth for overall enclosure depth of 11.30 inches (287.0 mm).
② Cover assembly consists of 2 side rails, top and bottom vented covers.
Important: Order individual device covers and blanks separately.

PRL4 Blank Covers

Used to cover blank space on chassis. All PRL4 cover heights are measured in "X" units. 1X equals 1.38 inches (35.1 mm).

Table 31. PRL4 Blank Covers

Cover Size	Style Number	
	24.00-Inch (609.6 mm) Width Box	36.00, 44.00-Inch (914.4, 1117.6 mm) Width Box
1X	6554C01H01	6554C02H01
2X	6554C01H02	6554C02H02
3X	6554C01H03	6554C02H03
4X	6554C01H13	6554C02H13
5X	6554C01H14	6554C02H14
6X	6554C01H04	6554C02H04
7X	6554C01H05	6554C02H05
9X	6554C01H06	6554C02H06
10X	6554C01H07	6554C02H07
11X	6554C01H08	6554C02H08
12X	6554C01H09	6554C02H09
13X	6554C01H10	6554C02H10
15X	6554C01H11	6554C02H11
20X	6554C01H12	6554C02H12



PRL4B Interior

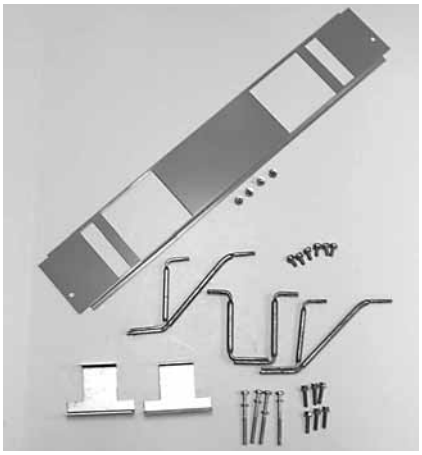
PRL4 Breaker Connector Kits

Breaker Connector Kits

Each kit includes copper connectors, mounting brackets, covers, hardware and instructions for mounting breaker(s) in a PRL4. **Breakers are not included.** Contact your local Satellite plant for availability and application information (see **Page 4**).

Connector Kit

Each kit includes copper connectors mounting brackets, cover, hardware and instructions.



Connector Kit

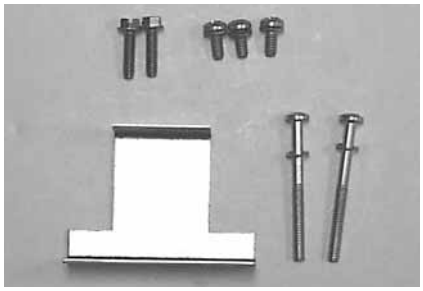
Table 32. Breaker Connector Kits

Breaker Frame	Space Required		Poles	Mounting Type	Connector Kit Catalog Number
	Inches (mm)	"X"			
EHD, FD, HFD EHD, FD, FDB, HFD, FDC ED, EDH, EDC	2.75 (69.9)	2X	1 ① 2 2	Twin Twin Twin	**KPRL4FD1** **KPRL4FD2** **KPRL4ED2**
EHD, FD, FDB, HFD, FDC FCL, FB-P, FD/LFD ED, EDH, EDC JD, JDB, HJD, JDC JD, JDB, HJD, JDC	4.13 (104.9)	3X	3 3 3 2, 3 2,3	Twin Twin Twin Single Twin	**KPRL4FD** **KPRL4FBP** **KPRL4ED** **KPRL4JDS** **KPRL4JDT** ②
DK, KD, KDB, HKD, KDC DK, KD, KDB, HKD, KDC CKD, CHKD	5.50 (139.7)	4X	2, 3 2,3 2,3	Single Twin Single	**KPRL4KDS** **KPRL4KDT** ③ **KPRL4CKD** ④
LCL LA-P LD, LDB, HLD, LDC, CLD, LC MDL, HMDL NB-P CND, CHND ND, HND	8.25 (209.5)	6X	2, 3 2, 3 2, 3 2, 3 2, 3 3 2, 3	Single Single Single Single Single Single Single	**KPRL4LCL** ② **KPRL4LAP** ② **KPRL4LD** ② **KPRL4MC** ② **KPRL4NBP** ③ **KPRL4CND** ③④ **KPRL4ND** ②

① Two sets of twin mounted 1-pole breakers.
② 36.00-inch (914.4 mm) minimum box width required.
③ 44.00-inch (1117.6 mm) box width required.
④ Requires density rated bus in existing panel chassis.

Hardware Kit

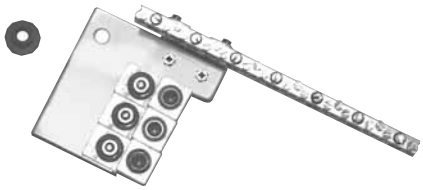
Each kit includes mounting bracket(s) and mounting hardware only. Use the appropriate Connector Kit catalog number and add an "H" to designate hardware only (example: KPRL4FD-H).



Hardware Kit

Standard Ground Bus

Copper bus with (3) 6 – 300 kcmil lugs plus a 24-circuit terminal bar with #14 – 1/0 wire range.



6572C746G01

PRL4 Fusible Connector Kits

Fusible Switch Connector Kits

Each kit includes copper connectors, extension wings (when required), hardware and instructions to mount a **fusible** switch. **Switches are not included.** Contact your local Satellite plant for availability and application information (see **Page 4**).

Table 33. Fusible Switch Connector Kits

Switch Height		Switch Ampere Rating	3-Pole Switch		Connector Kit
Inches (mm)	"X" Space Required		240 Volts	600 Volts	
			Catalog Number		
5.50 (139.7)	4X	30 – 30 60 – 60 100 – 100	**FDPWT3211R** **FDPWT3222R** **FDPWT3233R**	**FDPWT3611R** **FDPWT3622R** **—**	**—** **KPRL44X** ① **—**
6.88 (174.8)	5X	100 – 100	**—**	**FDPWT3633R**	**KPRL45X** ①
8.25 (209.6)	6X	200 200 – 200	**FDPBS324R** **FDPBT3244R**	**FDPBS364R** **FDPBT3644R**	**KPRL4B6XS** **KPRL4B6XT** ②
12.38 (314.5)	9X	400	**FDPW325R**	**FDPW365R**	**KPRL4W9X**
15.13 (384.3)	11X	600 800	**FDPW326R** **FDPW327**	**FDPW366R** **FDPW367**	**KPRL4W11X** **KPRL4W11X** ②
20.63 (524.0)	15X	1200	**FDPW328**	**FDPW368**	**KPRL4W15X** ②

① These connector kits will fit the FDP and FDPW switches.
② 44.00-inch (1117.6 mm) box width required for both R and J fuse applications.

PRL4 Breaker and Fusible Switch Retrofit Kits

Breaker Retrofit Kits

Each kit includes **one** breaker, copper connectors, covers, hardware and instructions to mount in a PRL4.

Table 34. Breaker Retrofit Kits

Breaker Frame	Frame Ampere Rating	Trip Range	Mounting Type
EHD	100	15 – 100	Twin
FDB		15 – 100	Twin
FD		15 – 100	Twin
HFD		15 – 100	Twin
FDC		15 – 100	Twin
FCL		15 – 100	Twin
FB-P		15 – 100	Twin
FDB	150	110 – 150	Twin
FD	225	110 – 225	Twin
HFD		110 – 225	Twin
FDC		110 – 225	Twin
ED		100 – 225	Twin
EDH		100 – 225	Twin
EDC		100 – 225	Twin
JD	250	70 – 250	Twin/Single
HJD		70 – 250	Twin/Single
JDC		70 – 250	Twin/Single
DK	400	100 – 400	Twin/Single
KD		100 – 400	Twin/Single
HKD		100 – 400	Twin/Single
KDC		100 – 400	Twin/Single
CKD		100 – 400	Single
LCL		125 – 400	Single
LA-P		70 – 400	Single
LD	600	300 – 600	Single
CLD		300 – 600	Single
HLD		300 – 600	Single
CHLD		300 – 600	Single
LDC		300 – 600	Single
CLDC		300 – 600	Single
MDL	800	300 – 800	Single
CMDL		300 – 800	Single
HMDL		300 – 800	Single
CHMDL		300 – 800	Single
ND	1200	600 – 1200	Single
CND		600 – 1200	Single
HND		600 – 1200	Single
CHND		600 – 1200	Single
NDC		600 – 1200	Single
CNDC		600 – 1200	Single

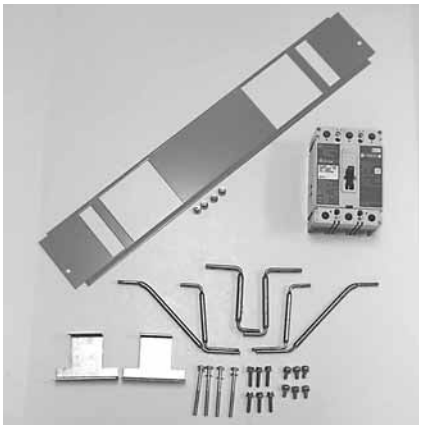
How to Order a Breaker Retrofit Kit by Catalog Number

Use "KPRL4" prefix and add catalog number of breaker as shown below. Use suffix "T" or "S" to denote twin or single mounting. Twin mounting indicates that one set of connectors is required to mount two breakers (of similar frames) opposite one another. RETROFIT KIT INCLUDES ONE BREAKER ONLY, FOR EITHER SINGLE OR TWIN MOUNTED APPLICATIONS.

Table 35. Catalog Numbering System — Breaker Retrofit Kit

KPRL4 KD 3 400 S

- **KPRL4** — Designates KPRL4 Connector Kit
- **KD** — Breaker Frame
- **3** — Number of Poles: **2** = 2-Pole, **3** = 3-Pole
- **400** — Ampere Rating
- **S** — **T** = Twin, **S** = Single



Breaker Retrofit Kit

Fusible Retrofit Kits

Each kit includes a 3-pole switch, copper connectors, extension wings (if required), hardware and instructions to horizontally mount in a PRL4.

Table 36. Fusible Retrofit Kits

Switch Ampere Rating	Switch Type	Mounting Type
30 – 30	FDPW	Twin
60 – 60	FDPW	Twin
100 – 100	FDPW	Twin
100	FDPW	Single
200	FDPB	Single
200 – 200	FDPB	Twin
400	FDPW	Single
600	FDPW	Single
800	FDPW	Single
1200	FDPW	Single

How to Order a Fusible Retrofit Kit by Catalog Number

Use "KPRL4" prefix and add catalog number of appropriate switch (refer to **Page 29** for 3-pole switch catalog number).

Example: The Retrofit Kit catalog number for a 600 volt, 100 ampere twin FDPW switch is:

Table 37. Catalog Numbering System — Fusible Retrofit Kit

KPRL4 FDPW T 3 6 33 R

- **KPRL4** — Designates PRL4 Connector
- **FDPW** — Switch Type: FDPW, FDPB
- **T** — **T** = Twin, **S** = Single
- **3** — **3** = 3-Pole
- **6** — Volts: **2** = 240, **6** = 600
- **33** — **11** = 30 – 30, **22** = 60 – 60, **33** = 100 – 100, **44** = 200 – 200, **4** = 200, **5** = 400, **6** = 600, **7** = 800, **8** = 1200
- **R** — **R** = R Fuse, **J** = J Fuse, **T** = T Fuse

PRL4 Energy Sentinel

Energy Sentinel

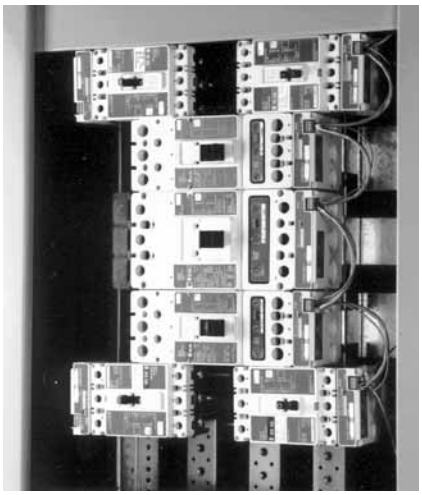
The IQ Energy Sentinel is a submetering device that mounts directly on a circuit breaker and monitors both power and energy with an overall accuracy of 99%.

This high system accuracy is achieved by use of the Cutler-Hammer SURE Plus Chip, which is a sophisticated microprocessor.

All that is necessary to complete an IQ Energy Sentinel installation is to insert it into the load side of a breaker, feed the load conductors through it, and run the shielded twisted pair wire for communications. The IQ Energy Sentinel has a nonvolatile memory, is powered by the circuit breaker, and can be applied on 3-phase, 4-wire or single-phase, 3-wire systems.

The space-saving design characteristics of the IQ Energy Sentinel mean they can be quickly and easily retrofitted onto Series C® circuit breakers in existing equipment…with no additional space required. Additionally, IQ Energy Sentinel can be installed when upgrading to Series C from older breakers that are physically interchangeable…with no additional space required.

Power and energy information from IQ Energy Sentinels can be communicated to a PC, a panel mounted Central Energy Display (CED), or even existing building management or distribution control systems.



Energy Sentinels Installed

Table 38. Energy Sentinel

Series C Breaker Frame	Voltage ac	Maximum Amperes	Catalog Number
F F	120/208, 120/240 277/480	150 150	**IQESF208** **IQESF480**
J J	120/208, 120/240 277/480	250 250	**IQESJ208** **IQESJ480**
K K	120/208, 120/240 277/480	350 350	**IQESK208** **IQESK480**

IQ Central Energy Display

IQ Central Energy Display

The IQ Central Energy Display may be panel mounted or located remotely (up to 7500 feet [2286 m] away). It displays power, peak demand, and energy readings of up to 50 IQ Energy Sentinels and eight IQ Data Plus meters.

Additional capabilities include: peak demand alarming, demand and energy totals for groups of IQ Energy Sentinels and IQ Data Plus 11 digital meters.

Refer to your local Satellite for retrofit and upgrade options available for existing equipment.

PRL1a, 2a, 3a Special Trims and Enclosures

Ventilated Trim

Required on PRL1a, 2a, 3a, 600 ampere and above panels only. Order by adding the letter "V" to the standard trim catalog number. **Add 10% to standard trim list price**.

Example: LT2072S becomes **LTV2072S**.

Ventilated Trim

Fastrim

Used when concealed trim mounting hardware is required for PRL1a, 2a and PRL3a. Trim clamps are included and shipped with the trim. Order by adding the letter "F" to the standard trim catalog number. **Add 20% to standard trim list price.**

Example: LT2072S becomes **LTF2072S**.

For trim clamps only, refer to **Page 20**.

Fastrim

Door-In-Door

Piano hinge on the right side of the trim provides access to the wiring gutters without requiring removal of the trim. Order by adding the letters "DD" to the standard trim catalog number. **Add 20% to standard trim list price.**

Example: LT2072S becomes **LTDD2072S**.

Door-in-Door

Type 12/3R Enclosures

The complete enclosure consists of a box and trim. The enclosure meets code requirements for both Type 12 (dust-tight) and Type 3R (rainproof) standards. Features include a laser cut trim with rounded corners, concealed hinges and a T-handle lock. Gasketing is provided around the trim door. The box is gasketed and made from code gauge steel with dripshield and is painted ANSI-61.

Table 39. Type 12/3R Enclosures for PRL1a, 2a, 3a

Box Dimensions in Inches (mm)			Catalog Number	
Height	**Width**	**Depth**	**Box**	**Trim**
24.00 (609.6) 36.00 (914.4) 48.00 (1219.2)	20.00 (508.0)	6.00 (152.4)	**VWPB2024** **VWPB2036** **VWPB2048**	**LWPT2024** **LWPT2036** **LWPT2048**
60.00 (1524.0) 72.00 (1828.8) 90.00 (2286.0)			**VWPB2060** **VWPB2072** **VWPB2090**	**LWPT2060** **LWPT2072** **LWPT2090**

Type 12/3R Enclosures

PRL4 Special Trims and Enclosures

Door-In-Door Trim

Door-In-Door Trim

A piano hinge on the right side of the trim provides access to the wiring gutter without requiring the removal of the trim. When used with a standard PRL4 box, a special mounting channel must be used to add extra depth to the enclosure.

An extra depth box, not requiring a mounting channel, is another available option. Contact your local Satellite for ordering information.

Table 40. Special Trims and Enclosures

Standard Box Catalog Number	Mounting Channel Style Number	Door-In-Door Trim Catalog Number	
		Surface	Flush
BX2457 BX2473 BX2490	8708C82G02 8708C82G03 8708C82G04	LDD2457STW LDD2473STW LDD2490STW	LDD2457FTW LDD2473FTW LDD2490FTW
BX3673 BX3690 BX4473 BX4490	8708C82G05 8708C82G06 8708C82G07 8708C82G08	LDD3673STW LDD3690STW LDD4473STW LDD4490STW	LDD3673FTW LDD3690FTW LDD4473FTW LDD4490FTW

Type 12/3R Enclosures

Type 12, 24.00 Inches (609.6 mm) Wide

Type 3R, 36.00 Inches (914.4 mm) Wide

PRL4 enclosures are available in both Type 12 (dust-tight) and Type 3R (rainproof) designs. The 24.00-inch (609.6 mm) wide enclosure includes a single hinged door while the 36.00-inch (914.4 mm) wide is provided with double hinged doors. The side gutter covers are an integral part of the box in all styles. Sizes and catalog numbers are shown in the table below.

Table 41. Type 12/3R Enclosures

Enclosure Dimensions in Inches (mm)			Catalog Number	
Height	Width	Depth	Type 3R	Type 12
57.00 (1447.8) 73.00 (1854.2) 90.00 (2286.0)	24.00 (609.6) 24.00 (609.6) 24.00 (609.6)	13.90 (353.1)	RPC2457 RPC2473 RPC2490	DPC2457 DPC2473 DPC2490
73.00 (1854.2) 90.00 (2286.0)	36.00 (914.4) 36.00 (914.4)		RPC3673 RPC3690	DPC3673 DPC3690

PRL5P Parts Section	*Page*
PRL5P Chassis Layout	**35**
PRL5P Breaker Adapter Unit Catalog Numbers	**36**
PRL5P Branch Breaker Information	**37**
PRL5P Main or Through-Feed Lugs	**38**
PRL5P Neutrals and Grounds	**39**
PRL5P Boxes, Trims and Filler Plates	**40**

Ordering Procedure

Step 1 Select the correct part or Branch Device. When selecting, you need to know the following:

- Panelboard type.
- Amperage.
- System voltage.
- Available short circuit rating.
- Number of poles available.
- Size and number of wires per phase.
- "X" space required.

Step 2 Refer to the 5P panelboard layout on **Page 35** to verify the amount of "X" space available.

Step 3 Create a 5P breaker unit catalog number, by following the instructions on **Page 36**, or order the catalog number for parts on **Pages 38** through **40**.

Step 4 Determine if extra filler covers are required. Additional filler covers may be necessary to fill the unused space. Refer to **Page 40** for filler plate information.

PRL5P Chassis Layout

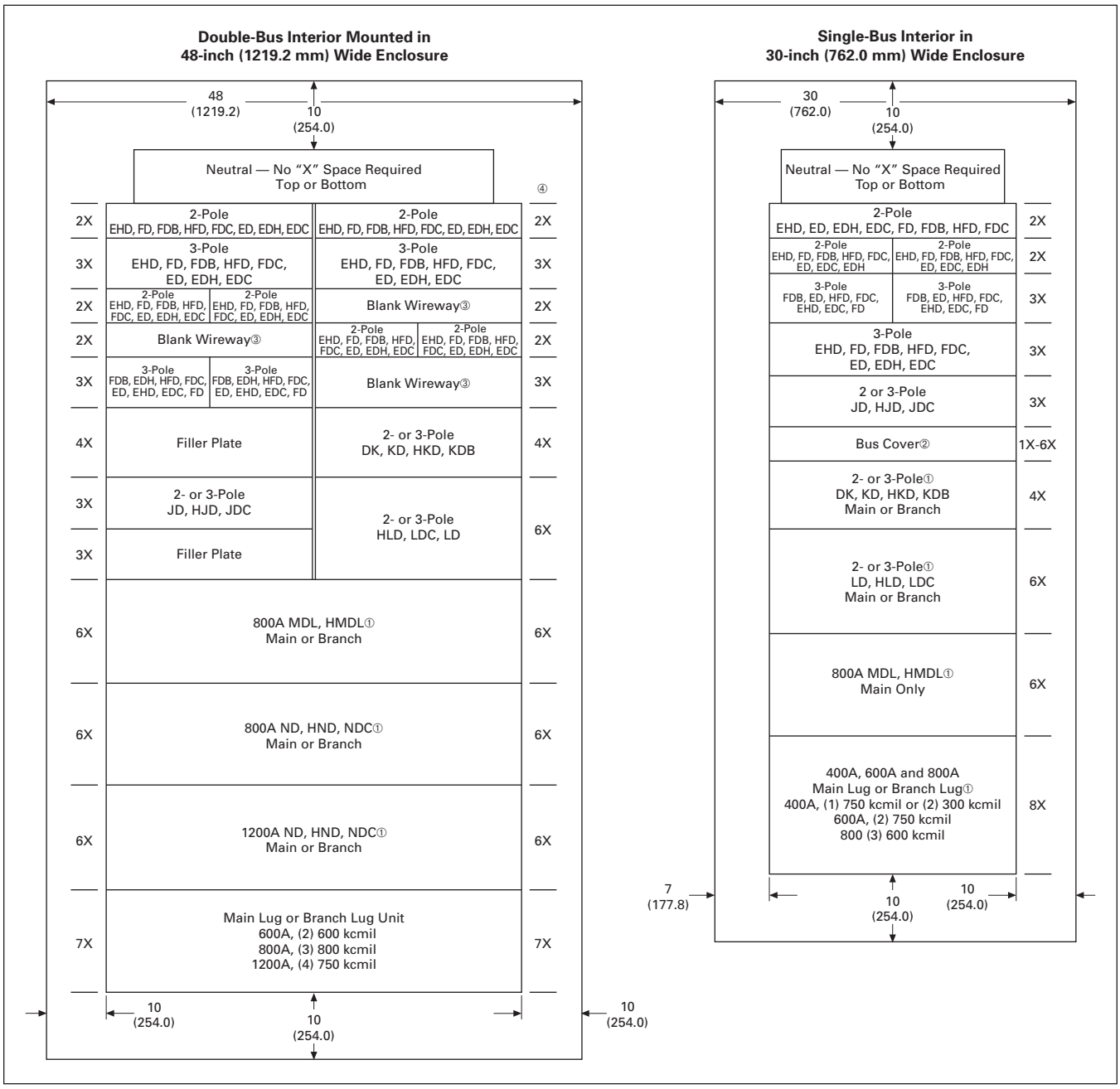
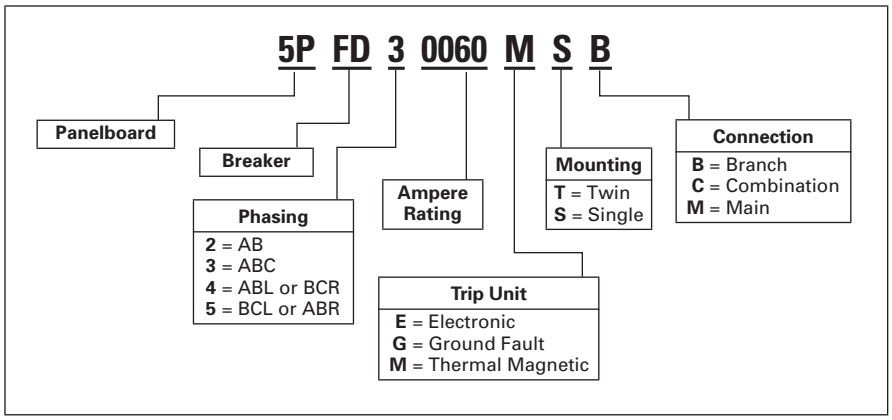


Figure 3. PRL5P Chassis Layout — Dimensions in Inches (mm)

① If used as a main device, must be mounted at the neutral end of panel.
② Fixed bus covers are required for unused spaces if NEC® six circuit disconnect rule is to be met.
③ Blank wireway fillers are required opposite any dual breaker unit.
④ One "X" = 1.38 inches (35.1 mm).

PRL5P Breaker Assemblies Catalog Numbers

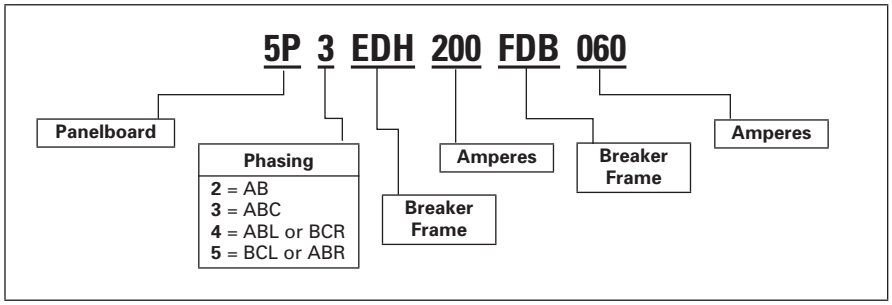
Table 42. Catalog Numbering System — 5P Single or Twin Breakers with Adapters



A plug-on unit is a complete assembly with a circuit breaker and mounting adapter to mount on a 5P panelboard.

Single indicates units that may be mounted in a single or double bus panel, and twin indicates double bus panels only. See **Tables 44** and **45** on **Page 37**.

Table 43. Catalog Numbering System — 5P Dual Breakers with Adapters



Any two F-Frame breakers listed may be mounted on the same 2X or 3X dual breaker adapter.

Dual breaker adapters may be used in single or double bus panels. Dual breaker adapters can **NOT** be mounted across from another adapter in a double bus panel. See **Table 46** on **Page 37**.

Branch Devices

Single-pole breakers in single adapter units. Include two or three single-pole 15 – 60 ampere assembled on one unit. (One X = 1.38 inches [35.1 mm])

Table 44. Single-Pole Breakers in Single Adapter Units

Breaker Type	Ampere Rating	Interrupting Rating (kA Sym.)				"X" Space Required
		120 Vac	240 Vac	277 Vac	125 Vdc	
EHD	15 – 60	—	—	14	10	2X
FD	15 – 60	—	—	25	10	2X
HFD	15 – 60	—	—	65	10	2X
EHD	15 – 60	—	—	14	10	3X
FD	15 – 60	—	—	25	10	3X
HFD	15 – 60	—	—	65	10	3X

Table 45. 2- and 3-Pole Breakers in Single Adapter Units

Breaker Type	Ampere Rating	Interrupting Rating (kA Sym.)				"X" Space Required
		240 Vac	480 Vac	600 Vac	250 Vdc	
ED	100 – 225	65	—	—	—	3X
EDH	100 – 225	100	—	—	—	3X
EDC	100 – 225	200	—	—	—	3X
EHD	15 – 60	18	14	—	10	3X
EHD	70 – 100	18	14	—	10	3X
FD	15 – 60	65	25	18	10	3X
FD	70 – 100	65	25	18	10	3X
FD	110 – 225	65	25	18	10	3X
HFD	15 – 60	100	65	25	22	3X
HFD	70 – 100	100	65	25	22	3X
HFD	110 – 225	100	65	25	22	3X
FDC	15 – 60	200	100	35	22	3X
FDC	70 – 100	200	100	35	22	3X
FDC	110 – 225	200	100	35	22	3X
JD, JDB	70 – 225	65	35	18	10	3X
JD, JDB	250	65	35	18	10	3X
HJD	70 – 225	100	65	25	22	3X
HJD	250	100	65	25	22	3X
JDC	70 – 225	200	100	35	22	3X
JDC	250	200	100	35	22	3X
DK	100 – 400	65	—	—	—	4X
KD, KDB	250 – 400	65	35	25	10	4X
HKD	250 – 400	100	65	35	22	4X
KDC	250 – 400	200	100	50	22	4X
LD, LDB	300 – 600	65	35	25	22	6X
HLD ①②	300 – 600	100	65	35	25	6X
LDC	300 – 600	200	100	50	25	6X
MDL ①②	400 – 800	65	50	25	22	6X
HMDL ①②	400 – 800	100	65	35	25	6X
ND	400 – 1200	65	50	25	—	6X
HND ①②	400 – 1200	100	65	35	—	6X
NDC	400 – 1200	200	100	50	—	6X

① For use only in double bus chassis panelboards.
② 100% rated breakers are **NOT** available in 5P panelboards.

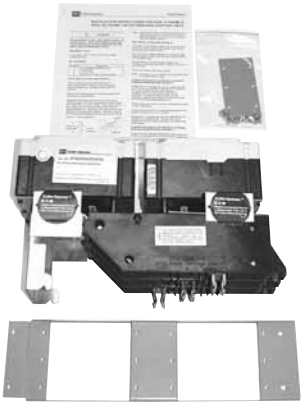
Dual breaker adapters — Any two breakers listed in **Table 46** may be mounted on the same 2X or 3X dual breaker adapter.

Dual breaker adapters may be used in single or double bus chassis. Dual breaker adapters can **NOT** be mounted across from another in a double bus chassis. (One X = 1.38 inches [35.1 mm])

Table 46. Dual Breaker Adapters

Breaker Type	Ampere Rating	Interrupting Rating (kA Sym.)				"X" Space Required
		240 Vac	480 Vac	600 Vac	250 Vdc	
ED	100 – 225	65	—	—	—	3X
EDH	100 – 225	100	—	—	—	3X
EDC	100 – 225	200	—	—	—	3X
EHD	15 – 60	18	14	—	10	3X
EHD	70 – 100	18	14	—	10	3X
FD	15 – 60	65	25	18	10	3X
FD	70 – 100	65	25	18	10	3X
FD	110 – 225	65	25	18	10	3X
HFD	15 – 60	100	65	25	22	3X
HFD	70 – 100	100	65	25	22	3X
HFD	110 – 225	100	65	25	22	3X
FDC	15 – 60	200	100	35	22	3X
FDC	70 – 100	200	100	35	22	3X
FDC	110 – 225	200	100	35	22	3X

5PFD30060MSB



5P3EDH200FDB060

PRL5P Main or Through-Feed Lugs

Table 47. PRL5P Main or Through-Feed Lugs

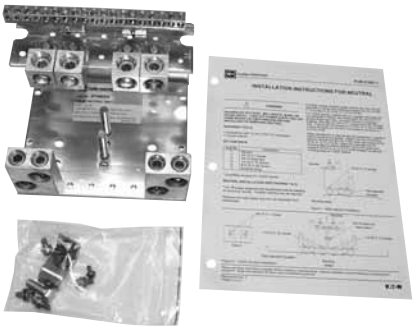
Description	Ampere Rating	Wire Size Range	"X" Space Required	Catalog Number
Single Bus Chassis Mounting				
Ampere Lug Unit	400	(1) 1/0 – 500 or (2) 1/0 – 250 kcmil	8X	**5PLUG3400SC**
Ampere Lug Unit	600	(2) 1/0 – 500 kcmil	8X	**5PLUG3600SC**
Ampere Lug Unit	800	(2) #2 – 500 or (3) #2 – 400 kcmil	8X	**5PLUG3800SC**
Double Bus Chassis Mounting				
1200 Ampere Lug Unit	600 – 1200	(4) #4 – 750 kcmil	7X	**5PLUG31200TC**

5PLUG3800SC

5PLUG31200TC

PRL5P Neutrals and Grounds

Neutral Assembly



5PN800A

Table 48. Neutral Assemblies with Lugs

Incoming Number of Cables and Wire Size	Catalog Number
(4) 250 – 500 kcmil 800 Amperes Aluminum/Copper	**5PN800A**
(4) 250 – 500 kcmil 800 Amperes Copper	**5PN800C**
(4) 250 – 500 kcmil 1200 Amperes Aluminum/Copper	**5PN1200A**
(4) 250 – 500 kcmil 1200 Amperes Copper	**5PN1200C**

Table 49. Additional Lugs for Neutral Assemblies

Description	Catalog Number
(1) 1/0 – 750 kcmil or (2) 1/0 – 300 kcmil Aluminum/Copper	**5PNL400**
(2) 250 – 500 kcmil Aluminum/Copper	**5PNL600**
(3) 3/0 – 750 kcmil Aluminum/Copper	**5PNL800**
(4) 3/0 – 750 kcmil Aluminum/Copper	**5PNL1200**
Ground Bar Type	
1200 Amperes Aluminum/Copper	**5PG1200A**
1200 Amperes Copper	**5PG1200C**

Ground Bar Assemblies

5PG1200A

Table 50. Grounded "B" Phase Adapter Kits

Ampere Rating	Main Device	Catalog Number
Single Bus Chassis		
400	Main Lugs	**5PCGBLUG400S**
600	Main Lugs	**5PCGBLUG600S**
600	LD Breaker	**5PCGBLD600S**
800	Main Lugs	**5PCGBLUG800S**
Double Bus Chassis		
800	MD Breaker	**5PCGBMD800T**
1200	Main Lugs	**5PCGBLUG1200T**
1200	ND Breaker	**5PCGBND1200T**

PRL5P Box, Trim and Deadfront Filler Plates

Table 51. PRL5P Box, Trim and Deadfront Filler Plates

Chassis "X" Factor	Catalog Number		
	Back Box	Trim	Trim Door Kit
Single Bus Chassis — 30.00-Inch (762.0 mm) Wide Box			
24X	**5PB2430G**	**5PT2430S**	**5PD24S**
32X	**5PB3230G**	**5PT3230S**	**5PD32S**
40X	**5PB4030G**	**5PT4030S**	**5PD40S**
Double Bus Chassis — 48.00-Inch (1219.2 mm) Wide Box			
24X	**5PB2448G**	**5PT2448S**	**5PD24T**
32X	**5PB3248G**	**5PT3248S**	**5PD32T**
40X	**5PB4048G**	**5PT4048S**	**5PD40T**

Table 52. Deadfront Filler Plates

Vertical "X" Increment	Catalog Number	
	Single Bus Chassis ①	Double Bus Chassis
1X	**5PFP1S**	**5PFP1T**
2X	**5PFP2S**	**5PFP2T**
3X	**5PFP3S**	**5PFP3T**
4X	**5PFP4S**	**5PFP4T**
5X	**5PFP5S**	**5PFP5T**
6X	**5PFP6S**	**5PFP6T**

① These fillers are also used across from a breaker unit in a double bus chassis.

5PT2430S

5PD24S

5PFP3S

5PFP3T

PRL1a, 2a-LX

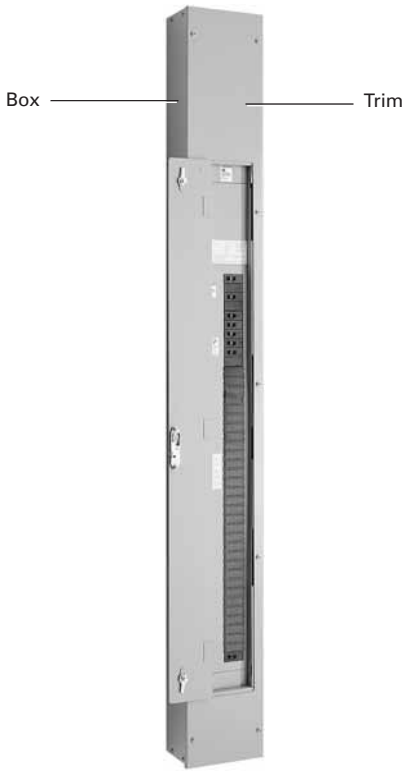


Table 53. Type 1 Box and Trims

Box Height in Inches (mm)	Catalog Number		
	Box	Surface Trim Standard	Surface Trim Door-In-Door
Incoming Location Top Fed			
69.00 (1752.6)	**YSC969**	**LTC969S**	**LTCD969S**
78.00 (1981.2)	**YSC978**	**LTC978S**	**LTCD978S**
81.00 (2057.4)	**YSC981**	**LTC981S**	**LTCD981S**
90.00 (2286.0)	**YSC990**	**LTC990S**	**LTCD990S**
Incoming Location Bottom Fed			
69.00 (1752.6)	**YSC969**	**LTC969SB**	**LTCD969SB**
78.00 (1981.2)	**YSC978**	**LTC978SB**	**LTCD978SB**
81.00 (2057.4)	**YSC981**	**LTC981SB**	**LTCD981SB**
90.00 (2286.0)	**YSC990**	**LTC990SB**	**LTCD990SB**

Pow-R-Command

For replacement parts, see PRL3a Section, **Page 21**. Parts available are the following:

- Connector kits.
- Ground assemblies.
- Service entrance kits.
- Deadfront covers.
- Trim locks.

Pow-R-Command

Additional Services

Since virtually all panelboards are supplied to meet specific customer requirements, other parts not listed in this publication might occasionally be needed. Price and availability for parts not shown here may be obtained by contacting your local Satellite plant and providing a complete description of the part along with the data on the panelboard nameplate.

Should you experience difficulty in determining what replacement parts are needed, contact your local Satellite Plant Manager who can provide help to:

- Identify and recommend replacement parts.
- Remove damaged parts and instruct you in how to install replacement parts.
- Verify the correct connector kits which should be ordered for each circuit breaker or fusible switch.
- Retrofit existing panelboard boxes with new Pow-R-Line interiors.
- Provide a recommended spare parts list.

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